

Reinforcement Learning for Stealthy Advertisement Placement in Dynamic Environments

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1 Project Overview

Unlike traditional literature that focuses on ad-detection (classification), this project models an autonomous agent (the Advertisement) that must navigate a web-page grid to maximize user engagement while dynamically evading a scanning Ad-Blocker. This represents a *Stochastic Stealth Pathfinding* problem in a non-stationary environment.

2 Environment Specification

The environment is a 10×10 discrete grid representing a website interface.

- **The Agent:** A 1×1 block representing a display ad.
- **The User Focus (The Target):** A 3×3 "Viewport" that moves vertically or horizontally every N steps, simulating a user scrolling or shifting attention. High rewards are only available inside this moving window.
- **The Ad-Blocker (The Threat):** The grid is divided into four 5×5 quadrants (Sectors). The Blocker cycles through these sectors. A "Warning Phase" (4 steps) precedes a "Scan Phase" (1 step). If the agent is in a sector during its Scan Phase, the episode terminates.

3 Markov Decision Process (MDP)

3.1 State Space (S)

The state is represented as a tuple $s = (x_a, y_a, x_t, y_t, \sigma, \tau)$:

- (x_a, y_a) : Current coordinates of the Ad agent.
- (x_t, y_t) : Center coordinates of the 3×3 User Viewport.
- $\sigma \in \{0, 1, 2, 3\}$: The Sector ID currently targeted by the Ad-Blocker.
- τ : Time remaining until the next Scan Phase (Cooldown/Warning timer).

3.2 Action Space (A)

Discrete action set: $\mathcal{A} = \{\text{Up, Down, Left, Right, Stay}\}$.

4 Research Objectives

1. **Trajectory Optimization:** Can the agent learn to "orbit" the user's focus while remaining outside the imminent danger sectors?
2. **Risk Sensitivity:** Observe how the agent prioritizes safety (staying far from the scanner) versus reward (staying in the viewport) as the scan timer τ approaches zero.
3. **Generalization:** Test if the agent, trained on linear scrolling, can adapt to erratic or "jumpy" user focus patterns.